HACKCANTON SEASON #2 · FINANCIAL APPLICATIONS

Tirai

The confidential multi-dealer RFQ / OTC desk, built native on Canton — settling in real CIP-56 assets: cETH, CBTC, Canton Coin, USDCx.

You whisper quotes. The market hears nothing.

github.com/PugarHuda/tirai · tirai.vercel.app · Pugar Huda Mantoro (solo)

THE TRADE THAT MOVES THE MARKET BEFORE IT HAPPENS

Asking for a price *is* the information

Request a quote on 50 million of a 30-year sovereign and you have told the market your size and your direction — before a single unit changes hands.

Trade in public

Public books and mempools leak size and direction. Front-running and adverse selection are structural, not incidental.

Trade off-chain

Phones and closed terminals leak nothing — and settle nothing. Counterparty risk, T+2, manual reconciliation.

That is why institutional block trading has not moved on-chain.

THE DEALER TERMINAL, ON THE LEDGER

Sealed quotes, atomic settlement, post-trade audit

1 · RFQ

Buyer asks a chosen dealer panel. The market never sees it.

2 · Sealed quote

Each dealer's price is delivered to the buyer alone — rivals never receive the contract.

3 · Escrow

Quoting locks the dealer's asset. A price is a commitment, not a bluff.

4 · Atomic DvP

Bond and cETH move in one transaction — or neither does.

5 · Report

The regulator sees executed trades only. Zero pre-trade visibility.

Three rails: **reverse-Vickrey** (second price, honest quoting is dominant), **direct bilateral OTC**, and **partial fills** on both.

NOT A UI PROMISE — A LEDGER PROPERTY

Dealer B's node never receives dealer A's quote

On Canton, privacy is **sub-transaction**: a contract is delivered only to its signatories and observers. There is no filtering, no encryption, nothing to misconfigure — the data simply never travels.

So the claim is falsifiable, and we falsify it on every run: an automated check queries the live ledger and asserts each dealer sees **only its own quotes** and the regulator sees **zero pre-trade contracts**. It exits non-zero if a single quote ever leaks.

CIP-56 CANTON TOKEN STANDARD

The cash leg is a real registry asset

Award creates a **TokenTrade** implementing the standard's **AllocationRequest** interface — so any standard wallet renders "*allocate 1,240,000 cETH to this settlement*" with no Tirai-specific integration. The registry locks the cash.

Settlement then runs in **one atomic transaction**: registry transfer of cETH to the winning dealer + escrowed bond delivered to the buyer + trade report to the regulator.

The cash instrument is any *{admin, id}* — **cETH, CBTC, Canton Coin and USDCx are one code path**, proven by a second registry in the test suite.

CONFIDENTIAL PRE-TRADE, PROVABLE POST-TRADE

Best execution without a public order book

A buyer — or a dealer defending its pricing — can disclose a single sealed quote to the regulator on demand. From those disclosures the desk proves the clearing price was at or below **every competing ask**.

Compliance gets better evidence than a public tape, because it includes the quotes that were never executed. Competitors get nothing.

LIVE ON CANTON DEVNET — VERIFIABLE RIGHT NOW

Not a mock-up

41 settled trades on-ledger + 5 atomic baskets	**16** best-execution attestations	**36** Daml test scripts green	**66** hosted QA checks, 3 browsers

~25 instruments across sovereigns, supranationals and corporates, on all three rails. Hosted read-only desk over live ledger state, plus a 6-tool MCP server so an evaluator can point their own agent at the desk.

THE FIFTH IMPLEMENTATION — AND THE FIRST NATIVE ONE

Four chains, four cryptography stacks, one ledger that needed none

BUILD	CHAIN	PRIVACY MACHINERY REQUIRED
Diam	Arbitrum (iExec)	TEE confidential compute, encrypted handles
Segel	Stellar (Soroban)	Two Groth16 circuits, hand-rolled Poseidon
Sealed Pair	Sui	Walrus commitments + Seal threshold encryption
Samar	Ethereum (Zama)	FHE, branchless encrypted settlement
Tirai	Canton	None. Sub-transaction privacy is the ledger model.

HOW IT MAKES MONEY, AND WHO BUYS

A venue fee the contract collects itself

Revenue

Basis points of notional in the settlement asset, taken atomically inside the settlement transaction — no invoicing, no collection risk. Plus CIP-0047 activity markers accruing network rewards on every trade.

Customer

Desks whose tickets are large enough to leak ($1m–$100m), in an asset already tokenised, with an obligation to evidence best execution. Beachhead: crypto-native desks trading cETH and CBTC.

Distribution through hosting venues (Temple, Bron, Console, Canton Loop) that already hold the client relationship and custody — Tirai ships as an embedded app and shares the fee.

WHAT IS DONE, WHAT IS NOT

Built and verified · pending and stated

Done

Model, three settlement rails, CIP-56 allocation path against the real Splice interfaces, privacy asserted on the live network, hosted desk, MCP server, 36 tests, full seeded audit trail.

Not yet

No live cETH/CBTC transaction — blocked on the test-token grant. No paying customer, no design partner signed. Deployed on the shared 5N validator; hackcanton-01 needs participant-admin rights (DAR upload 403).

The ask: cETH and CBTC Devnet test tokens, an introduction to one hosting venue, and one design-partner desk.

TIRAI — INDONESIAN FOR "CURTAIN"

Price discovery happens behind it.

You whisper quotes. The market hears nothing — and the regulator still gets the whole record.

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